Polling Pops deployment architecture guide

Polling Pops uses a **modular monolith deployed on one virtual machine**, supported by external services. Its business features form one coordinated application, while separate containers run the website, API, database and supporting tools.

This guide brings together the account checklist, provider boundaries and five diagrams for a real deployment rehearsal. AWS EC2 is the selected staging provider in the repository. Cloudflare, Resend, Better Stack and S3 in a separate backup account are the proposed supporting products. These diagrams describe the intended connections; they do not establish that any account, integration or environment has been provisioned.

Start with staging. Staging and production each need their own VM, credentials, database and live media. They can use the same hosting provider account while remaining separate deployments.

Products and architectural roles

Product	Architectural role	What it provides
GitHub and GitHub Container Registry	Development and delivery	Source code, independent review, automated checks and signed release images
AWS EC2 and EBS	Hosting infrastructure	A Linux virtual machine and persistent disk for the running application
Cloudflare Registrar and DNS	Domain and DNS	Domain ownership and lookup of the application's server address
Resend	External email service	Verification links and sign-in PIN delivery
Better Stack	Observability and incident response	Logs, metrics, traces, external uptime checks and incident notifications
Separate AWS account and S3	Disaster recovery	Encrypted recovery copies outside the active server's account
Stripe	External payment service	Hosted checkout, subscriptions, billing portal and payment events

PostgreSQL, Valkey, MinIO, ClamAV and Caddy run in the Docker stack. They do not require separate hosted-service accounts. The application is organized into presentation, application and data layers, all deployed on the same VM for this topology.

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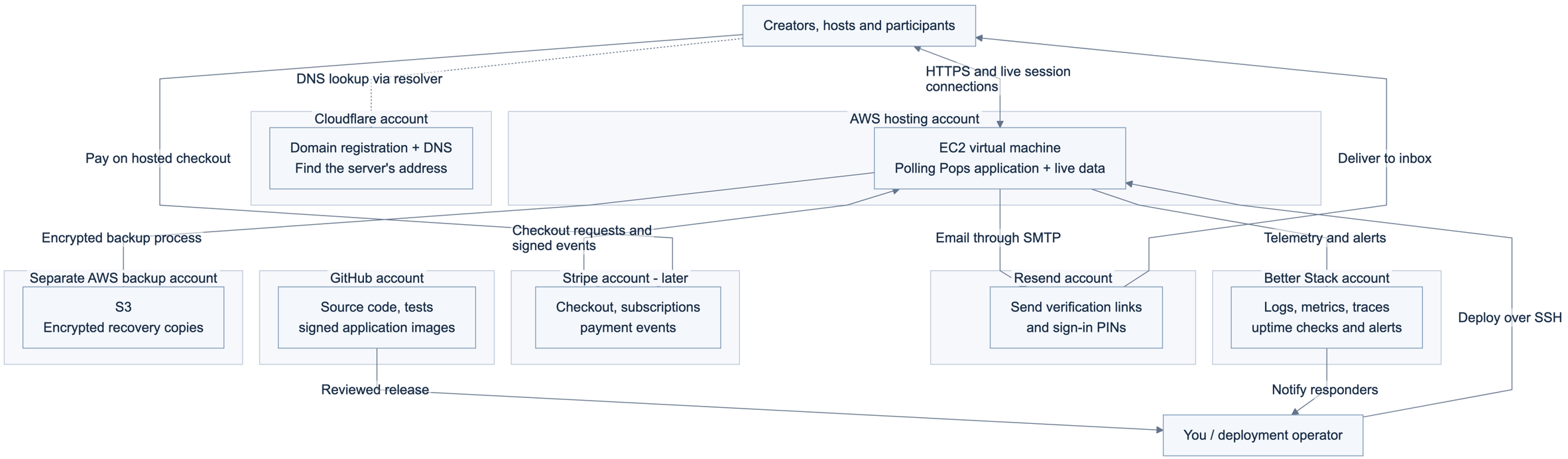
Account checklist in priority order

- ☐ **1. GitHub:** reuse the existing account with administration access to `riojung/pollingpops`. Arrange a second human collaborator with their own account and repository write access for independent review. Configure the protected staging environment before the first signed-image build.
- ☐ **2. AWS hosting:** create or reuse an account with billing enabled, MFA and a budget alert. Prepare EC2 and EBS access. The documented starting host is Linux **amd64**, with **4 vCPUs, 8 GB RAM and 100 GB SSD**. Launch it when the required deployment inputs are ready.
- ☐ **3. Cloudflare Registrar and DNS:** create an account and register an available domain, or use an existing domain and DNS provider. One domain can provide separate application and media subdomains for staging and production. The diagrams below use Cloudflare in DNS-only mode.
- ☐ **4. Resend:** create an account, verify the sending domain through DNS and create a dedicated API key for SMTP. Prepare a test inbox for verification links and sign-in PINs.
- ☐ **5. Better Stack:** create one account for Telemetry and Uptime/Incidents. Configure the telemetry source, separate host log shipping, uptime checks and distinct page, warning and ticket alert integrations. The readiness observer adapters also need provider-specific configuration.
- ☐ **6. AWS backup account and S3:** create a separately administered backup account and a private bucket with encryption, versioning and retention. Keep backup credentials and recovery keys outside the active application VM. Configure scheduled backup jobs and a clean-host restore drill.
- ☐ **7. Stripe:** create an account when billing testing begins and use its test environment first. Stripe can wait for the initial billing-disabled staging deployment. The unchanged production configuration and release gates require a real Stripe test-mode billing rehearsal; complete merchant and bank setup before accepting live payments.

Items 1-6 supply the accounts for the staging deployment and recovery exercises. Account creation must be followed by configuration, integration and verification. Public production promotion also requires the existing release-readiness gates.

1. Overall product and account boundaries

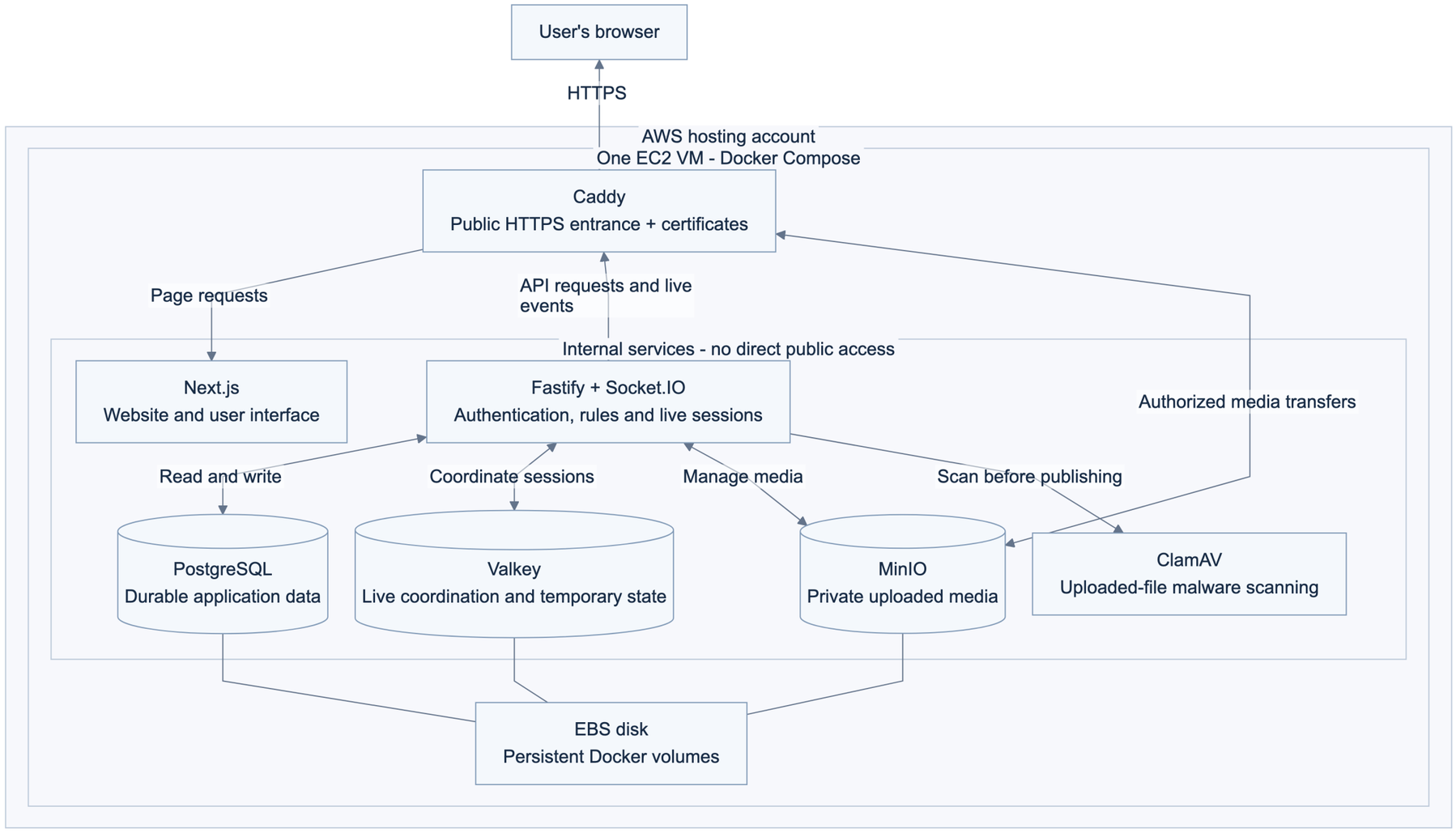
Each outer box represents a provider or account boundary. Solid arrows show application, deployment or operational flows. The dotted arrow represents DNS lookup.



Cloudflare in **DNS-only mode** tells browsers where the server lives; browsers then connect directly to AWS. Enabling Cloudflare's proxy would add Cloudflare to the web traffic path and requires a separate reviewed ingress configuration. See Cloudflare proxy status.

2. Inside the AWS virtual machine

EC2 is the rented computer. Docker Compose runs the components together on that computer. Caddy is the public HTTPS entrance; the other services communicate over internal networks.



The three logical application layers are:

- **Presentation:** Next.js serves pages and the browser user interface.

- **Application:** Fastify owns authorization, business rules and API operations; Socket.IO carries live session updates.

- **Data:** PostgreSQL stores durable application records and MinIO stores private media. Valkey supports live coordination and is rebuildable.

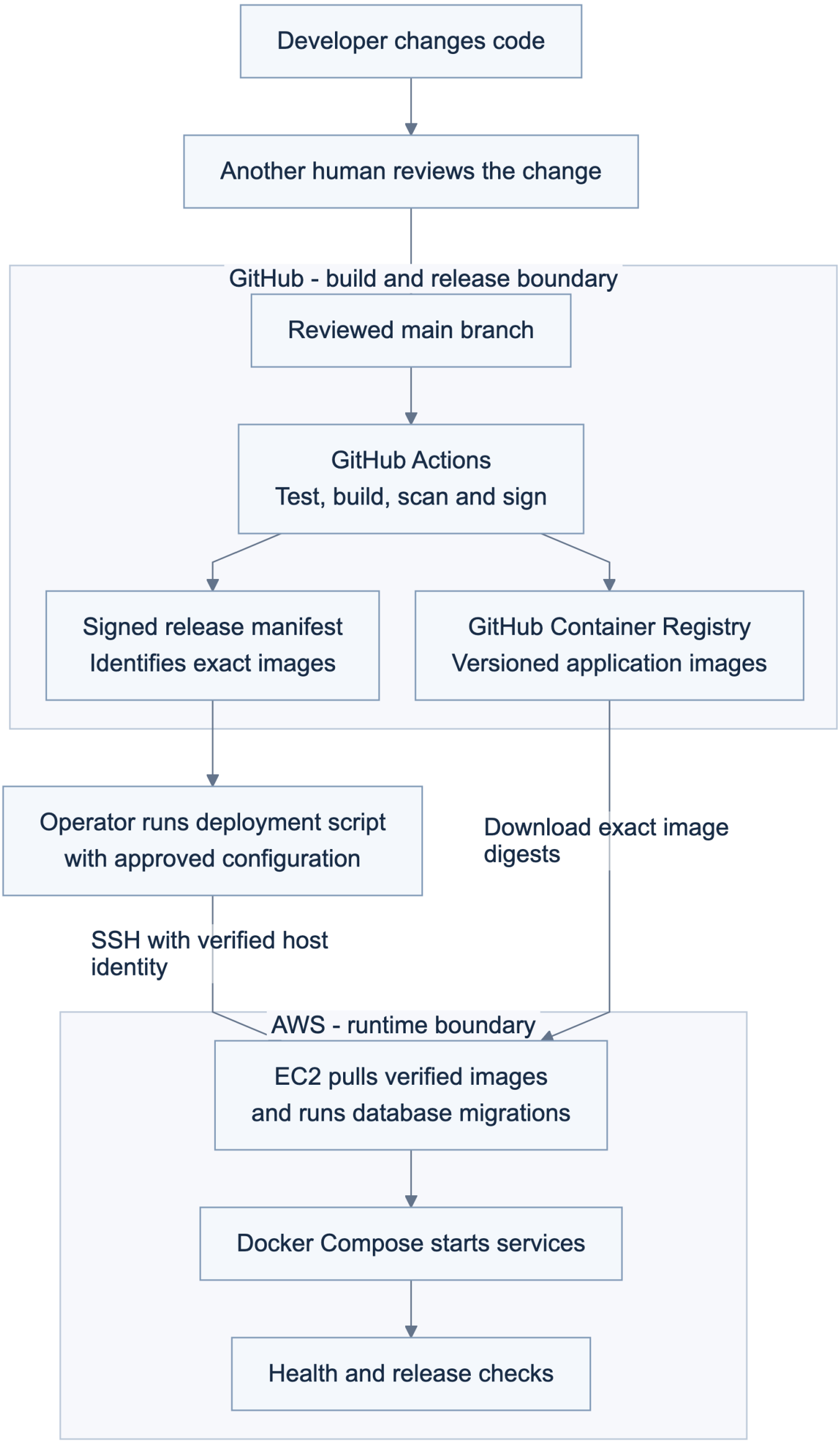
PostgreSQL is the durable source of truth. The API authorizes media access and issues constrained, short-lived upload or download URLs. Uploads enter MinIO quarantine; the server validates and scans the content before promoting clean files.

Caddy routes the app hostname to Next.js and the API, and the separate media hostname to MinIO. The public media route does not make the bucket anonymous. Database, cache, scanning, metrics and administration endpoints remain private.

All these containers share one VM and disk failure domain. Losing that VM interrupts the service; container restart policies do not provide failover. The minimum host size is a starting point that requires capacity testing on the actual target.

3. From code to a running deployment

GitHub prepares the release. The operator initiates deployment to AWS using the reviewed release manifest and configuration.



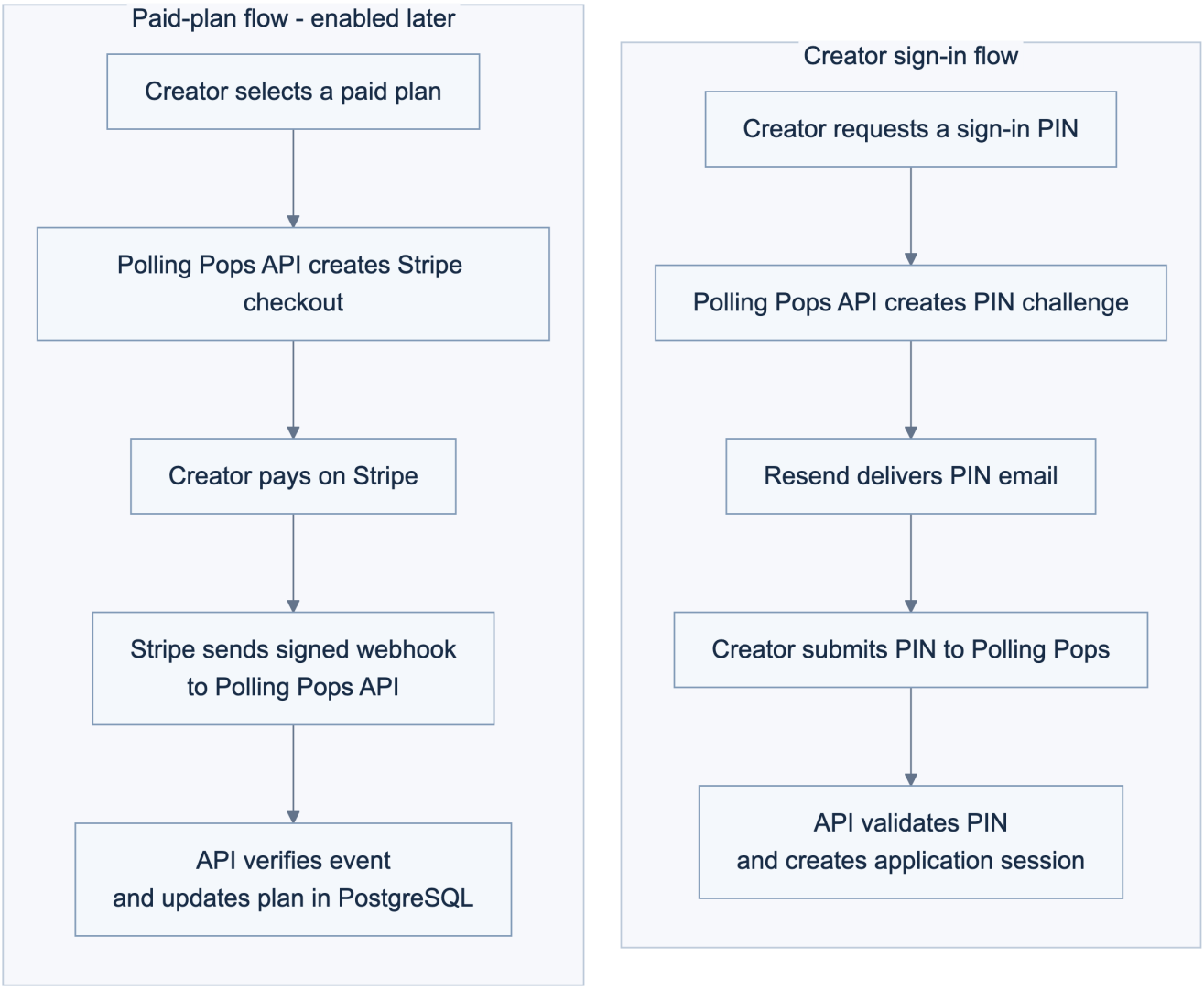
This is the **CI/CD architecture**. Continuous integration checks changes; the controlled deployment process installs a reviewed release. GitHub Actions signs release images using its approved identity. GitHub Container Registry stores those images, so no separate container-registry provider account is needed.

The documented workflow uses the operator's deployment script over SSH. The diagram does not imply that GitHub Actions automatically logs into the production server. Runtime database credentials remain restricted; the one-shot migration receives separate elevated credentials.

Production adds the protected release-tag, signed-release and readiness acceptance requirements in the deployment runbook.

4. Email authentication and payments

These are separate workflows. Both refer to the same Polling Pops API running on EC2.



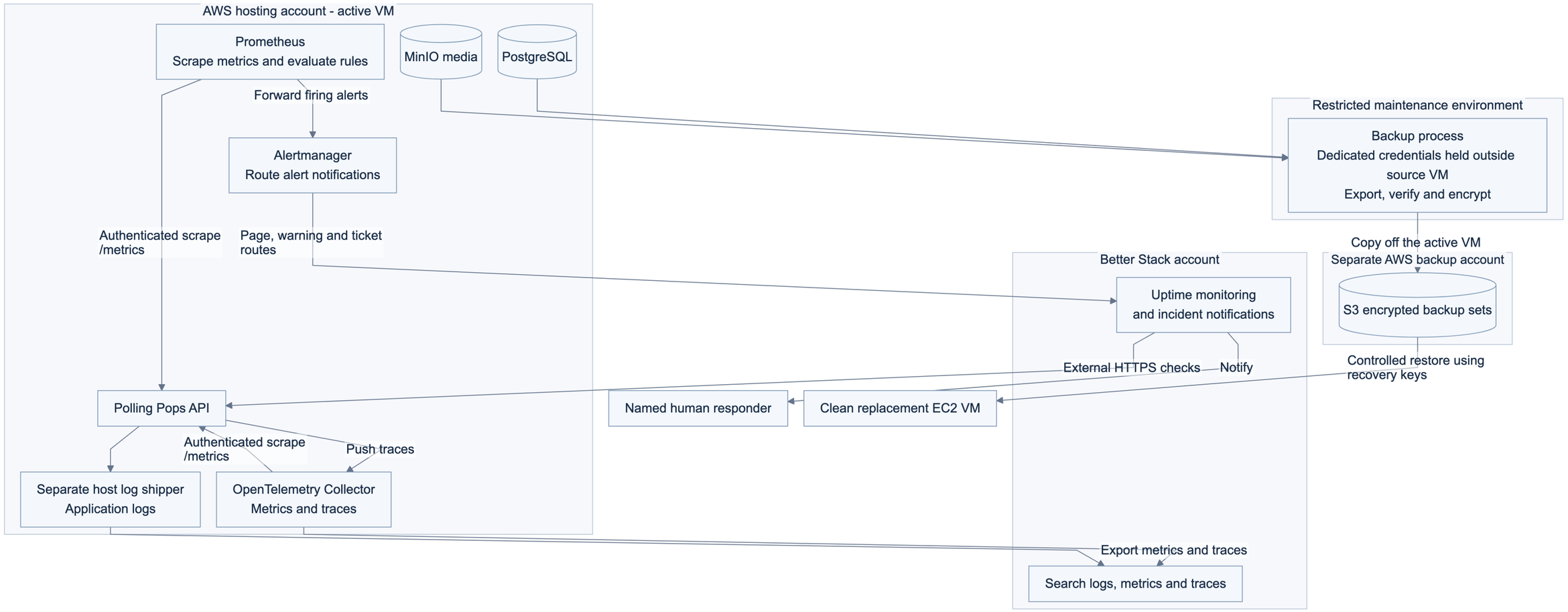
New creators first verify their email address through a separate verification link. Verification establishes mailbox ownership; it does not sign them in. They then request a six-digit PIN to create an application session. Participants can join sessions without creating accounts.

Resend delivers the messages through SMTP; Polling Pops owns identity verification, PIN validation and sessions. Domain verification records live in DNS.

Stripe owns checkout and payment processing; Polling Pops owns application permissions. Stripe webhooks enter through Caddy. The API verifies their signatures and handles event identity and ordering before changing entitlements. Returning to a browser success page does not itself grant an upgrade. Billing can remain disabled during the initial staging exercise.

5. Monitoring and disaster recovery

Monitoring reports what is happening. Backups provide recoverable copies of durable data.



The API pushes traces to the private OpenTelemetry Collector. The collector and Prometheus each initiate authenticated scrapes of the API's `/metrics` endpoint. The scrape arrows show request initiation; metric responses travel back over those connections. Prometheus evaluates alert rules and forwards firing alerts to Alertmanager, which routes page, warning and ticket notifications to distinct receiver integrations. A separately managed host log shipper sends application logs; the supplied telemetry overlay does not collect Docker logs itself. Better Stack also checks the public HTTPS endpoint from outside the VM.

Better Stack supports the required OpenTelemetry ingestion and Alertmanager webhooks. Configuring these connections, log shipping, readiness observers and named responders remains deployment work.

MinIO serves **live media**. S3 holds **backup copies** of the PostgreSQL export and media, including the metadata and tags required for recovery. Backups must be encrypted, verified and retained outside the active VM; recovery keys must remain available outside it as well. The backup process runs from a restricted maintenance environment using dedicated source-read identities and a separate destination-write identity. Backup credentials stay outside the source VM and the backup payload; source access uses approved restricted connections. S3 versioning helps retain previous object versions, but a successful clean-host restore is the recovery exercise.

The replacement VM is a restore target, not an automatically running standby. The current single-VM topology does not provide automatic failover or a high-availability guarantee.

Environment and access boundaries

Boundary	Rule
Staging and production	Separate VMs, credentials, databases, media and Compose projects
Internet and application	Public web and media requests enter through Caddy; operator SSH is restricted
Application and database	The long-running application uses a restricted role; migrations use separate credentials
Public media route and private storage	Short-lived authorized transfers; quarantine and scanning before publication
Hosting and backup accounts	Separate administration and credentials, with encrypted off-host copies
Application and external providers	Dedicated credentials for email, telemetry, alerts and payments
Operator and independent reviewer	A second human reviews the protected release path

Operational references

- Architecture and protocol
- Single-VM hosting profile
- Beta release preparation
- Build, service and deployment runbook
- Staging readiness and observer configuration
- Repository governance
- Backup and restore
- API reference including billing
- Production deployment configuration
- Release-readiness ledger